

Deep Learning for Extreme Weather Forecasting: Architectures, Benchmarks, and Frontiers

Abstract

The integration of deep learning (DL) into meteorology has precipitated a paradigm shift from traditional Numerical Weather Prediction (NWP) to data-driven Artificial Intelligence Weather Prediction (AIWP). While AIWP models have demonstrated proficiency in global medium-range forecasting, their application to extreme weather events—characterized by non-Gaussian distributions, rare occurrences, and high societal impact—remains a formidable challenge. This survey provides a comprehensive synthesis of 300 recent academic contributions focusing on deep learning methodologies for extreme weather forecasting. We categorize the field into distinct methodological families: global foundation models, generative and diffusion-based approaches, physics-informed hybrid architectures, probabilistic uncertainty quantification, and specialized nowcasting systems. Our analysis reveals that while AI models often outperform NWP in synoptic-scale pattern recognition, they systematically underestimate the intensity of record-breaking extremes due to regression-to-the-mean biases inherent in standard loss functions. We examine emerging solutions, including asymmetric loss functions, diffusion-based refinement, and hybrid spectral nudging techniques that recover high-frequency details. Furthermore, we critically evaluate the limitations of current benchmarks, the challenges of out-of-distribution generalization in a warming climate, and the nascent field of agentic reasoning for diagnostic analysis. This survey underscores that the future of extreme weather forecasting lies not in replacing physical models but in developing robust, hybrid, and generative frameworks capable of capturing the tail risks of the atmospheric distribution.

1. Introduction

Weather forecasting has historically been the domain of physics-based Numerical Weather Prediction (NWP), which solves discretized partial differential equations governing atmospheric dynamics. However, the computational cost of high-resolution ensembles and the inherent uncertainties in parameterizing sub-grid processes have motivated the exploration of data-driven alternatives. The advent of deep learning, particularly Transformer architectures, Graph Neural Networks (GNNs), and Neural Operators, has enabled the development of AIWP models that can emulate atmospheric evolution with orders-of-magnitude lower inference latency [1, 2, 3]. Models such as Pangu-Weather, GraphCast, and FourCastNet have achieved skill levels comparable to or exceeding operational NWP systems like the ECMWF Integrated Forecasting System (IFS) for standard metrics such as geopotential height and temperature at medium ranges [1, 4, 2].

Despite these successes, the forecasting of extreme weather events—tropical cyclones, heatwaves, atmospheric rivers, and convective storms—presents unique challenges that standard AIWP models struggle to address. Extreme events reside in the tails of the probability distribution, often representing out-of-distribution (OOD) samples relative to the historical reanalysis data used for training. A recurring finding across recent literature is that AI models, optimized primarily using Mean Squared Error (MSE) or similar L_2 norms, tend to produce “smooth” forecasts that regress toward the climatological mean, thereby underestimating the intensity of extreme phenomena [5, 3, 6]. For instance, AI models have been shown to lag behind traditional NWP in predicting sudden-turning typhoons and record-breaking heatwaves, often failing to capture the sharp gradients and peak intensities associated with high-impact events [6, 5].

The motivation for this survey stems from the urgent need to bridge the gap between the general

proficiency of AIWP and the specific demands of extreme event prediction. As climate change increases the frequency and severity of extreme weather, the ability to accurately forecast these tails becomes critical for disaster risk reduction and infrastructure resilience. Recent research has diverged into several promising directions: the development of specialized loss functions that penalize extreme errors more heavily [7, 8]; the adoption of generative models, particularly Diffusion Models, to recover high-frequency spatial details and generate realistic ensemble spreads [9, 10]; and the creation of hybrid physics-AI frameworks that constrain data-driven predictions with physical conservation laws [11, 12].

This survey synthesizes evidence from 300 relevant academic papers published primarily between 2020 and 2026. We organize the discourse into a coherent taxonomy of methodological systems. Section 2 details the core methodologies, ranging from global foundation models and their limitations in extreme regimes to advanced generative and physics-informed architectures. Section 3 addresses the critical issues of evaluation, benchmarking, and uncertainty quantification, highlighting the discrepancies between reanalysis-based verification and ground-truth observations. Finally, Section 4 discusses the open challenges, including extrapolation limits in a changing climate, the “gray swan” problem, and the emerging role of agentic frameworks in meteorological reasoning. By integrating these diverse strands of research, we aim to provide a definitive roadmap for the next generation of extreme weather forecasting systems.

2. Method

The methodological landscape of deep learning for extreme weather forecasting is diverse, encompassing architectures ranging from global autoregressive transformers to local convolutional nowcasters and hybrid physics-informed networks. This section organizes these approaches into coherent families, analyzing their technical mechanisms, innovations, and specific efficacy in capturing extreme phenomena.

2.1 Global Foundation Models and Autoregressive Architectures

The backbone of modern AI weather prediction consists of large-scale foundation models trained on decades of reanalysis data (e.g., ERA5). These models typically employ autoregressive rollouts to predict future atmospheric states.

2.1.1 Transformer and Graph-Based Global Models Leading architectures include Pangu-Weather, which utilizes a 3D Earth-Specific Transformer (3DEST) to process atmospheric variables as cubic data, incorporating pressure levels as a spatial dimension to model inter-level relationships [1]. Similarly, GraphCast employs a Graph Neural Network (GNN) on an icosahedral mesh to propagate information across the globe, achieving state-of-the-art performance on standard metrics [4]. FourCastNet leverages Adaptive Fourier Neural Operators (AFNO) within a Vision Transformer backbone, utilizing frequency-domain token mixing to reduce spatial complexity to $\mathcal{O}(N \log N)$, enabling high-resolution (0.25°) global forecasting [2].

While these models excel in capturing large-scale synoptic patterns, they exhibit systematic deficiencies in extreme regimes. Comparative analyses reveal that AIWP models often underestimate the intensity of extreme wind speeds and precipitation quantiles. For example, during Storm Ciarán, models like FourCastNet and Pangu-Weather accurately captured the cyclone’s track and upper-level structure but systematically underestimated peak surface wind speeds in frontal-fracture regions compared to IFS HRES [13]. Similarly, in the context of tropical cyclones, AI models

demonstrate a “smoothing” effect, leading to errors in predicting sudden turns where local core interactions dominate over large-scale steering flows [6]. The identification of the Western Pacific Subtropical High (WPSH) strength as a key determinant for AIWP performance divergence highlights that these models rely heavily on large-scale steering flows and fail when local dynamics prevail [6].

2.1.2 Limitations in Record-Breaking Extremes A critical limitation of global foundation models is their inability to extrapolate beyond the training distribution. When evaluated on record-breaking events (e.g., the 2020 heat records or the 2024 Dubai rainfall), AI models systematically underpredict heat and wind intensity while overpredicting cold records, with errors growing linearly relative to the margin of record exceedance [5]. This behavior is attributed to the spectral bias of neural networks and the dominance of MSE loss functions, which penalize large deviations heavily, encouraging the model to output conservative, mean-like predictions [14, 3]. Studies on the 2021 Pacific Northwest heatwave indicate that while AI models can match local accuracy, they often fail spatially and temporally, lacking the structural realism to capture the persistence and magnitude of such compound events [15, 16].

Furthermore, long-term autoregressive rollouts suffer from instability regimes categorized as “blow-up,” “drift,” or “loss of seasonality.” Although models like Aurora and SFNO show stability over hundreds of days, they tend to suppress high-frequency energy amplification, effectively acting as denoisers that smooth out extreme perturbations rather than amplifying them as required for storm development [17]. This smoothing is particularly detrimental for predicting the intensification of tropical cyclones, where AI models often fail to extrapolate from weaker storms to unseen Category 5 events if such extremes were underrepresented or removed from the training set [18].

2.2 Generative Models and Diffusion-Based Refinement

To address the smoothing bias inherent in deterministic regression models, the field has increasingly turned to generative architectures, particularly Diffusion Models and Generative Adversarial Networks (GANs), which can recover high-frequency details and generate probabilistic ensembles.

2.2.1 Diffusion Models for Extreme Value Recovery Diffusion models have emerged as a powerful tool for refining coarse AI forecasts. FuXi-Extreme integrates a frozen FuXi autoregressive backbone with a conditional Denoising Diffusion Probabilistic Model (DDPM) to restore fine-scale details in surface variables like precipitation and wind speed [9]. By learning to reverse Gaussian noise added to ground truth data, the diffusion component effectively recovers the high-frequency spatial details lost during the autoregressive smoothing process. Similarly, ExtremeCast employs a cascaded framework where a deterministic Swin-Transformer is refined by a conditional diffusion model, utilizing an asymmetric loss function (Exloss) to correct the underestimation bias typical of MSE optimization [8].

The efficacy of diffusion models extends to ensemble forecasting. GenCast, a diffusion-based ensemble model, generates sharp, realistic sample trajectories rather than blurred ensemble means, outperforming the ECMWF ensemble on 97.4% of targets, particularly at lead times greater than 36 hours [10]. By modeling the conditional probability distribution $p(X^{t+1}|X^t)$ via conditional diffusion, these models can compute non-linear functionals of the forecast distribution more accurately than deterministic counterparts. In the context of downscaling, Anemoui-D2 utilizes a graph transformer-based diffusion model to predict high-resolution residuals conditioned on coarse inputs,

preserving large-scale dynamics while stochastically generating physically consistent small-scale extremes [19].

2.2.2 GANs and Adversarial Training Generative Adversarial Networks (GANs) have also been applied to enhance extreme event prediction. Kunyu, a global weather model, integrates adversarial losses alongside supervised regression to mitigate forecast blurring, learning distribution characteristics rather than just pixel-wise averages [20]. This approach sharpens forecast details and improves extreme event estimation. In precipitation nowcasting, GA-SmaAt-GNet employs a conditional GAN with an attention-augmented discriminator to reduce underestimation in extreme rainfall events by incorporating binary precipitation masks as explicit spatial priors [21].

However, generative models are not without limitations. While diffusion models achieve superior visual fidelity, they can systematically underestimate extreme wind intensities due to over-generalization [22]. Additionally, the computational cost of iterative denoising remains a barrier for operational deployment, with some models requiring significantly more inference time than deterministic baselines [10, 23]. Despite these challenges, the consensus is that generative approaches are essential for capturing the multi-modal nature of extreme weather distributions, which deterministic models often collapse into a single mean trajectory.

2.3 Physics-Informed and Hybrid Architectures

Recognizing the limitations of purely data-driven approaches, a significant body of work focuses on hybridizing AI with physical constraints, either through architectural design, loss function regularization, or coupling with NWP systems.

2.3.1 Hybrid ML-Physics Coupling via Spectral Nudging One of the most effective strategies for improving extreme event forecasting is spectral nudging, where large-scale fields from an AI model are assimilated into a high-resolution NWP model. The FuXi-SHTM framework integrates FuXi’s large-scale forecasts into the SHTM (WRF-based) mesoscale model via spectral nudging, correcting large-scale dynamics while retaining mesoscale structure fidelity [12, 24]. This hybrid approach has demonstrated a 16.5% reduction in typhoon track forecast error at 72 hours compared to standalone AI models [12]. Similarly, Pangu SP applies spectral nudging to constrain WRF with Pangu-Weather forecasts, effectively combining Pangu’s accurate large-scale circulation with WRF’s mesoscale intensification capabilities, resulting in a 20.3% reduction in average track error for Typhoon Doksuri [11].

These hybrid systems leverage the strengths of both paradigms: the AI model provides accurate boundary conditions and large-scale steering flows, while the physical model resolves fine-scale processes like convection and terrain interactions that are critical for extreme intensity prediction. However, challenges remain in tuning the nudging coefficients and cutoff wavelengths to prevent the suppression of legitimate mesoscale variability [11].

2.3.2 Physics-Informed Neural Networks (PINNs) and Constraints Another approach involves embedding physical laws directly into the neural network architecture or loss function. NowcastNet integrates a stochastic generative network with a deterministic evolution network governed by modified 2D continuity equations, adaptively combining mesoscale patterns dictated by physical conservation laws with data-driven learning [25]. This physics-embedded approach significantly outperforms pure NWP and optical flow methods in detecting extreme precipitation hotspots.

Other methods enforce specific physical constraints, such as mass and energy conservation. FuXi with hybrid sigma-pressure coordinates and conservation schemes enforces global dry air mass and moisture budget constraints, improving near-surface wind representation over complex terrain [26]. Similarly, LUPIN (Lagrangian Double U-Net) employs a fully differentiable semi-Lagrangian extrapolation operator integrated within the computational graph, enforcing divergence-free constraints to ensure physically consistent advection [27]. While these methods improve physical consistency, they often face trade-offs in computational efficiency and may struggle to capture highly nonlinear atmospheric interactions that deviate from the simplified physical assumptions embedded in the model [28, 29].

2.3.3 Neural Operators and Equation Discovery Neural operators, such as Fourier Neural Operators (FNO) and DeepONets, offer a resolution-invariant framework for learning solution operators of PDEs. FourCastNet’s use of AFNO allows for efficient global modeling, while newer architectures like OneForecast utilize multi-scale graph structures with dynamic gating units to capture high-frequency features essential for extreme disturbances [30, 2]. Additionally, research into interpretable equation discovery, such as the GEV-Informed Interpretable Equation Discovery method, uses PCA and Generalized Extreme Value (GEV) transformations to derive linear models that capture wind gust extremes with reduced geographical bias [28]. These approaches aim to bridge the gap between black-box deep learning and interpretable physical dynamics.

2.4 Specialized Models for Specific Extreme Phenomena

Given the diversity of extreme weather, specialized models tailored to specific phenomena (e.g., tropical cyclones, heatwaves, precipitation) often outperform general-purpose foundation models.

2.4.1 Tropical Cyclone Tracking and Intensity Tropical cyclone (TC) forecasting requires precise modeling of vortex dynamics and intensity changes. Storm-centric post-processing frameworks, such as those extracting $60^\circ \times 60^\circ$ storm-centered fields from global NeWM outputs, have shown superior probabilistic forecasting skills compared to using ERA5 initial conditions alone [31]. These frameworks bypass storm detection algorithms to prevent full misses and employ lead-time-adaptive spatial masking to isolate relevant features.

For intensity prediction, hybrid models like TIFNet utilize non-iterative spatiotemporal transformers to generate 0-120h intensity trajectories in a single forward pass, reducing error accumulation and achieving a 43% MAE reduction in rapid intensification scenarios compared to ECMWF IFS [32]. However, AI models still struggle with sudden-turning typhoons and often exhibit a low intensity bias, particularly for Category 5 storms, due to the smoothing effects of global RMSE cost functions and the scarcity of such events in training data [6, 18]. To address this, methods like the “Gray Swan Factory” use differentiable weather models and gradient descent optimization to manufacture specific extreme event scenarios, revealing that small perturbations to extratropical flow fields can drive transitions to extreme outcomes [33].

2.4.2 Heatwave and Temperature Extremes Heatwave forecasting benefits from custom loss functions that emphasize tail events. ExtNet employs a symmetric exponential loss function targeting both positive and negative temperature extremes, significantly improving heat wave prediction skill without degrading general performance [7]. Similarly, Targeted Concept Tuning (TaCT) uses sparse autoencoders to disentangle hidden representations into mono-semantic concepts, allowing for targeted intervention on failure modes related to typhoons and heatwaves [34].

Studies on the 2021 Pacific Northwest heatwave reveal that deep learning sensitivity analysis via backpropagation can optimize initial conditions up to 23 days in advance, revealing finite-amplitude synoptic errors rather than small-scale noise as the primary source of unpredictability [35]. However, AI models often exhibit persistent cold biases prior to heat wave onset, and their skill degrades significantly for subseasonal temperature extremes [36, 16].

2.4.3 Precipitation Nowcasting and Convective Initiation Precipitation nowcasting (0-6 hours) is critical for flash flood warnings. Models like MetNet and MetNet-2 utilize axial self-attention and large context neural networks to emulate atmospheric physics directly from data, outperforming HRRR for lead times up to 12 hours [37, 38]. Recent advances include FuXi-Nowcast, which couples large-scale 3D atmospheric states from ML weather models with local observations to predict convective initiation, addressing the longstanding challenge of predicting storm onset without NWP latency [39].

Generative models are particularly prevalent in nowcasting. CasCast decouples deterministic mesoscale prediction from probabilistic small-scale generation using a frame-wise guided diffusion transformer, achieving a 91.8% improvement in CSI for regional extreme precipitation [40]. Similarly, StormDiT utilizes a unified generative latent diffusion model with Rectified Flow to bridge the 2-6 hour “gray zone,” outperforming baselines by 140% in heavy rain CSI [41]. Despite these advances, models still struggle with the precise localization of convective initiation and often exhibit spatial blurring or false alarms in complex terrain [39, 25].

2.5 Uncertainty Quantification and Ensemble Methods

Reliable extreme weather forecasting requires robust uncertainty quantification (UQ). Traditional ensemble methods are computationally expensive, prompting the development of AI-specific UQ techniques.

2.5.1 Input Perturbation and Huge Ensembles Flow-dependent input perturbation methods, such as Hemispheric-Centered Bred Vectors (HCBV) and Huge Ensembles (HENS), generate diverse ensemble members by perturbing initial conditions fed into deterministic AI models [42, 43]. HENS, which combines bred vector perturbations with multi-checkpoint model uncertainty, has generated 7,424-member ensembles that sample extreme surface temperatures exceeding both observations and GEV-extrapolated NWP bounds [43]. These methods reveal that AI models can capture temperature extremes better than precipitation extremes, where epistemic uncertainty dominates [42].

2.5.2 Probabilistic Architectures and Evidential Learning Architectures like SwinVRNN integrate variational perturbation modules to learn multivariate Gaussian distributions for feature perturbation, enabling dynamic adjustment of noise intensity across spatial locations [44]. Evidential Deep Learning (EDL) extends spatio-temporal transformers to output Normal-Inverse-Gamma distribution parameters, deriving uncertainty directly from a single forward pass, which is significantly faster than ensemble or Monte Carlo Dropout methods [45].

Post-processing techniques also play a crucial role. GEV-Constrained Post-Processing hierarchies predict location, scale, and shape parameters of Generalized Extreme Value distributions for wind gusts, yielding superior probabilistic forecasts compared to direct NWP ensembles [46]. Similarly, Power Mean Ensemble Aggregation assigns higher weights to ensemble members predicting extremes, effectively shifting decision boundaries to improve extreme event detection [47].

2.6 Agentic Frameworks and Diagnostic Reasoning

A nascent but promising direction is the use of Large Language Models (LLMs) and agentic frameworks for diagnostic reasoning and automated analysis. The Extreme Weather Expert (EWE) framework operationalizes expert workflows via knowledge-enhanced planning and self-evolving closed-loop reasoning, utilizing dual-auditor modules to verify code correctness and visualization clarity [48]. Such systems can decompose complex multi-scale analysis tasks and generate structured reports, although their performance relies heavily on the quality of initial expert annotations.

Multimodal Large Language Models (MLLMs) like MeteorPred ingest 4D meteorological tensors alongside textual warnings to predict severe weather events, leveraging dynamic temporal gating and text-driven spatial masking to align grid data with query contexts [49]. While these models show promise in integrating diverse data modalities, they currently struggle with fine-grained subordinate categories and causal reasoning without task-specific fine-tuning [50, 49].

3. Challenges and Outlook

Despite significant progress, deep learning for extreme weather forecasting faces profound challenges that must be addressed to ensure operational reliability and societal benefit.

3.1 Extrapolation and the “Gray Swan” Problem

A primary concern is the ability of AI models to generalize to out-of-distribution (OOD) events, often termed “gray swans”—events that are physically possible but absent from the training data. Studies demonstrate that when extreme samples are removed from training sets, models like Four-CastNet fail to extrapolate, producing false negatives for unprecedented Category 5 storms [18]. Similarly, AI models exhibit a “temporal lag,” effectively predicting climates from 15-20 years prior due to regression to the mean of their training data, leading to cold biases in modern warming scenarios [51].

The distinction between *translocation* (learning from globally similar events) and *extrapolation* (learning from locally weaker events) is critical. GraphCast has shown success in forecasting regional gray swans via translocation enabled by global effective receptive fields, whereas models with limited receptive fields fail to capture these dynamics [14]. However, the systematic underestimation of precipitation extremes due to spectral bias remains a pervasive issue, limiting the utility of AI models for unprecedented rainfall events like the 2024 Dubai floods [14].

3.2 Evaluation Bottlenecks and Benchmark Discrepancies

Current evaluation practices often rely on reanalysis data (e.g., ERA5) as ground truth, which itself is a smoothed model product. This can obscure biases in AI models. The Extreme Weather Bench (EWB) initiative advocates for verification against ground-based observations (e.g., GHCNh, IBTrACS), revealing that global error metrics often hide poor performance on specific high-impact hazards [52]. For instance, AI models may outperform HRES in global metrics but struggle with long-lead time predictions for freeze events or exhibit warm biases in heatwaves when validated against station data [52].

Furthermore, standard metrics like RMSE and ACC are insensitive to extreme events. Threshold-weighted scoring rules (e.g., twCRPS) and event-specific metrics (e.g., SEDI for extremes) are necessary to properly evaluate model skill in the tails of the distribution [53, 54]. The lack of

standardized benchmarks for extreme events has led to fragmented evaluation protocols, hindering direct comparison between methodologies [55, 56].

3.3 Physical Consistency and Safety Concerns

Purely data-driven models often violate physical conservation laws (mass, energy, momentum), leading to unrealistic forecasts over long lead times. While hybrid models mitigate this, the integration of physical constraints often comes at the cost of predictive skill or computational efficiency [26, 57]. Fine-scale noise in raw deep learning outputs can disrupt geostrophic and continuity balances, generating erroneous spatial gradients that compromise the physical realism of the forecast [57].

Safety and robustness are also critical. Adversarial attacks, though theoretical, have shown that small perturbations to initial conditions can trigger catastrophic forecast failures, such as generating fake hurricane wind speeds with minimal noise injection [58]. Additionally, the “black-box” nature of deep learning models hinders trust and interpretability, making it difficult for forecasters to diagnose failure modes or understand the physical mechanisms driving a prediction [59, 60]. Explainable AI (XAI) methods like SHAP and Integrated Gradients are being explored to identify precursors and validate model reasoning, but causal guarantees remain elusive [60, 61].

3.4 Deployment and Operational Integration

Transitioning from research to operations involves significant hurdles. The computational cost of generative models, particularly diffusion-based approaches, can be prohibitive for real-time forecasting [10, 23]. Moreover, the dependency on high-quality reanalysis data for training creates a bottleneck, as reanalysis products are updated with significant latency. Real-world deployment requires models to handle near-real-time operational data, which often contains biases and gaps not present in curated reanalysis datasets [56, 62].

Infrastructure resilience also demands downscaled, local forecasts. While global models provide excellent synoptic guidance, they lack the resolution to capture local extremes driven by topography or urban heat islands. Hybrid correction-downscaling frameworks that decouple storm-scale bias correction from terrain-aware downscaling are emerging as a solution, but their generalizability across diverse geographical regions remains to be proven [63, 30].

3.5 Future Research Directions

Future research should prioritize the development of **hybrid foundation models** that seamlessly integrate physical solvers with data-driven components, ensuring conservation laws are satisfied by design [64, 65]. **Generative ensemble methods** must be optimized for efficiency to provide reliable probabilistic forecasts of extremes without excessive computational overhead [10, 66]. **Causal discovery** and **mechanistic interpretability** should be deepened to move beyond correlation-based predictions, enabling models to understand the physical drivers of extreme events [67, 68].

Furthermore, the community must invest in **climate-adaptive training strategies**, such as data augmentation with climate model projections or fine-tuning on recent warming trends, to ensure models remain robust in a non-stationary climate [51, 69]. Finally, the integration of **agentic AI** for automated diagnostic reasoning and decision support could transform how forecasters interact with AI systems, turning raw predictions into actionable insights for disaster management [48, 70].

4. Conclusion

Deep learning has irrevocably altered the landscape of weather forecasting, offering unprecedented speed and competitive accuracy for global medium-range predictions. However, the forecasting of extreme weather events remains a frontier where purely data-driven approaches face significant limitations, primarily due to smoothing biases, extrapolation failures, and physical inconsistencies. This survey has highlighted that the path forward lies not in discarding physical models but in synergizing them with advanced AI architectures.

Generative models, particularly diffusion-based refinements, offer a promising avenue for recovering the high-frequency details essential for extreme event intensity. Hybrid frameworks employing spectral nudging and physics-informed losses demonstrate that constraining AI with physical laws can significantly improve track and intensity forecasts for tropical cyclones and convective storms. Moreover, the development of specialized evaluation benchmarks and uncertainty quantification methods is critical to accurately assess model performance in the tails of the distribution.

As the climate continues to warm, the frequency and severity of extreme events will increase, demanding forecasting systems that are not only accurate but also robust to out-of-distribution scenarios. The integration of causal reasoning, agentic diagnostics, and climate-adaptive training will be paramount in building the next generation of AI weather models. By addressing the current challenges of physical consistency, extrapolation, and evaluation, the meteorological community can harness the full potential of deep learning to safeguard lives and infrastructure against the growing threats of extreme weather.

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